

SYSTEM ARCHITECTURE OVERVIEW

In various embodiments, the disclosed system comprises a plurality of sensing nodes and a computing system configured to communicate with the nodes, store measurements, derive features and metrics, and present privacy-preserving outputs. The nodes may operate independently or cooperatively, and may be used in any order or combination depending on the workflow, user preference, or clinical or consumer context.

A. Sensing Nodes

In certain embodiments, the system includes a first node configured for all-day and/or night measurement of external tumescence, rigidity, and physiology. The first node may be implemented as a partially open ring or a fully enclosed ring and may include one or more sensors comprising temperature sensing, pressure sensing, strain sensing, magnetic sensing using a hall-effect sensor and magnet, optical pulse sensing, inertial sensing, and combinations thereof.

In certain embodiments, the system includes a second node configured as an intravaginal longitudinal physiology monitor for extended wear. The second node may include one or more sensors comprising bio-impedance sensing, multi-frequency impedance sensing, capacitive wetness or dielectric sensing, tissue hydration sensing, solid-state pH sensing including ISFET-based pH sensing, temperature sensing, pelvic activity sensing, distributed pressure sensing, capacitive pressure sensing, piezoresistive sensing, MEMS sensing, and motion/orientation sensing including inertial sensing for artifact rejection.

In certain embodiments, the system includes a third node configured as a self-scan external geometry measurement device. The third node may be implemented as a partially open ring or a fully enclosed ring and may include one or more sensors comprising optical flow sensing for length determination, time-of-flight sensing, lidar, or infrared sensing for diameter determination, inertial sensing for artifact rejection, and optional anti-spoof sensing including optical pulse sensing and contact sensing.

In certain embodiments, the system includes a fourth node configured as an intraluminal geometry, compliance, and contact device configured for use intravaginally and/or anally. The fourth node may include one or more endpoint sensors including infrared time-of-flight sensors, and may include a controlled radial stimulus mechanism including an annular inflatable cuff, microbladder ring, constraint band, micro-pump, manual pneumatic actuation, mechanical cam, or combinations thereof. The fourth node may include one or more stimulus pressure sensors comprising MEMS pressure transducers and one or more contact sensing elements including circumferential pressure rings, piezoresistive or FSR rings, capacitive pressure rings, piezoresistive film bands, capacitive proximity/contact confirmation electrodes, temperature sensing, and inertial sensing for robustness, spoof resistance, and artifact rejection.

In certain embodiments, the system includes a fifth node configured as an upper-body physiologic context device, which may be implemented as a chest-mounted device, a ring, or a wrist-worn device. The fifth node may include sensors comprising electrical cardiac sensing using ECG electrodes, optical pulse sensing, inertial sensing, respiratory dynamics sensing or respiratory proxy sensing, temperature or thermal context sensing, electrodermal activity sensing, skin conductance sensing, contact integrity sensing, and wear detection sensing.

B. Computing System and Communications

The computing system may include one or more processors and memory configured to receive data from the nodes using wired or wireless communication, including Bluetooth, Bluetooth Low Energy, Wi-Fi, near-field communication, or other communication methods. In some embodiments, one or more nodes store data locally and upload the data periodically. In some embodiments, the computing system operates in real time to provide on-device processing, and in other embodiments the computing system performs batch processing or cloud-assisted processing.

C. Outputs and Privacy-Preserving Presentation

The computing system may generate privacy-preserving outputs comprising abstract indices, ranges, zones, curves, trend lines, and confidence indicators. Outputs may be computed within a node, within a mobile device, within a cloud service, or distributed across components. In various embodiments, the system avoids explicit anatomical imagery and avoids explicit anatomical labeling while still enabling meaningful longitudinal pattern measurement, cross-device correlation, and cross-user correlation.